Lost in Translation

LingoQuest

Software Engineering Project

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Semester: Spring 2026

Group Number: 9

Coordinator: Harry

Name of the Guide: Dr. Tushara Sadasivuni

Date: 04/06/2026

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1.0 INTRODUCTION

1.1 Software Engineers' information

1. **Lasya Jonnalagadda**

I'm an honors computer science major graduating in May 2026. I have skills in software engineering and data engineering from my Co-Op at State Street Bank. I participated in many full-stack, machine-learning, and cloud technology projects and helped build award-winning, scalable, and impactful solutions.

2. **Marquez Johnson**
I'm a computer science student graduating in December 2026. I've gained skills with full-stack projects of my own personal portfolio website, chatbot, and competing in a group KSU colorstack peach hacks hackathon last year. I utilized api, group organization, and presentation skills in order to present for the hackathon.
3. **Harry Ahsan**
A computer science senior planning to graduate soon. I have taken multiple classes and participated in projects that utilized full-stack operations and software dev methodologies to create functional programs and applications.
4. **Faysal Abdulkadir**
I'm a computer science student graduating in May 2026 with strong experience in front-end development. I've built several projects, including my personal portfolio website, a chess game, Connect 4, and other small web applications. While my primary strength is front-end, I've also explored back-end development and am continuing to build my skills in that area.

1.2 Planning and Scheduling (Sprint 2)

Assignee Name	Email address	Task	Duration (hours)	Dependency	Due Date	Evaluation
Lasya Jonnalagadda	Ljonnalagadda1@student.gsu.edu	Problem Statement Class Diagram Architectural Modeling	4 hours	None	04/06/2026	100 %
Marquez Johnson	mjohnson428@student.gsu.edu	Implementation	4 hours	Database management (Lasya)	04/06/2026	100 %
Harry Ahsan	tahsan1@student.gsu.edu	UML Sequence Diagram Use Case Diagram Test Cases	4 hours	None	04/06/2026	100 %
Faysal Abdulkadir	fabdulkadir2@student.gsu.edu	Context Diagram Activity Diagram Behavioral Modeling	4 hours	None	04/06/2026	100 %

1.3 Teamwork Basics

Two things get accomplished in good teams: the task gets accomplished, and the satisfaction of team members is high.

Ground Rules:

- Work Norms: Work will be distributed based on skill levels and interest to take on. First, we will note down all tasks and allow members to decide what they want to work on. If some tasks are left behind, we will give those to members who have the capacity to take them on. For the majority, we will keep the teamwork transparent, but if members have different work habits, we will try to find a middle ground. We can peer review the work so that we don’t miss any details.
- Facilitator Norms: As a group, we will rotate the position of the facilitator who will focus on the tasks that need to be done, make sure all members are participating, keep the team on track to meet deadlines, help the team resolve problems, summarize the members’ decision and keep the team aware of what’s happening.
- Communication Norms: As a team, we will communicate either in-person or over the phone. This way, we will be easy to reach each other and not lead to any confusion.
- Meeting Norms: Meeting either before or after class times is most feasible to meet in person; if we are to meet virtually, evenings would work better. If someone doesn’t show up to a meeting even after agreeing to show up, they will have to make sure their assigned work is completed to standard. If the teammate informed us that they can’t attend, that’s understandable, so we will try to find a time that works for everyone.
- Consideration Norms: The same rules as our class will remain at our meetings. If anyone is dominating the discussion, the facilitator can ask everyone to calm down and take turns discussing again. If these norms have to be changed, anyone can bring it up, and we can discuss if it’s okay with everyone, and we can change the norms.

Handling Difficult Behavior

- We will simply discuss the issue at hand as a group and resolve the issue by talking.

Handling Group Problems

- We will email the professor if we have any group problems, whether it be technical or person related.

1.4 (Revised) Problem Statement (Sprint 4)

Our product is an AI-powered educational language learning web platform designed to help users overcome language barriers through adaptive learning, pronunciation testing, and AI assistance. The system includes

1. A comprehensive user management module for registration and secure login.
2. The system provides secure user and admin logins
3. AI-based skill assessment pre-tests
4. Level-based learning modules
5. Adaptive question difficulty
6. Pronunciation evaluation
7. Digital sketchpads note-taking
8. Integrated AI chatbot support

Unlike existing platforms such as Duolingo and Babbel, our system integrates

9. Real-time AI personalization
10. Voice analysis
11. UI customization features and persistent user preferences.
12. To maintain engagement, the system includes:
    - Email notifications, reminders, and weekly newsletters.
    - Visual progress tracking via a streak system and progress bars.
    - Shareable milestone awards and auditory feedback for task completion.

Client-requested features such as:

1. Background color customization
2. Mouse icon personalization
3. Secure preference persistence has been incorporated into the system.

The project is compelling due to its global applicability, AI integration, and full-stack implementation, making it both technically innovative and socially impactful.

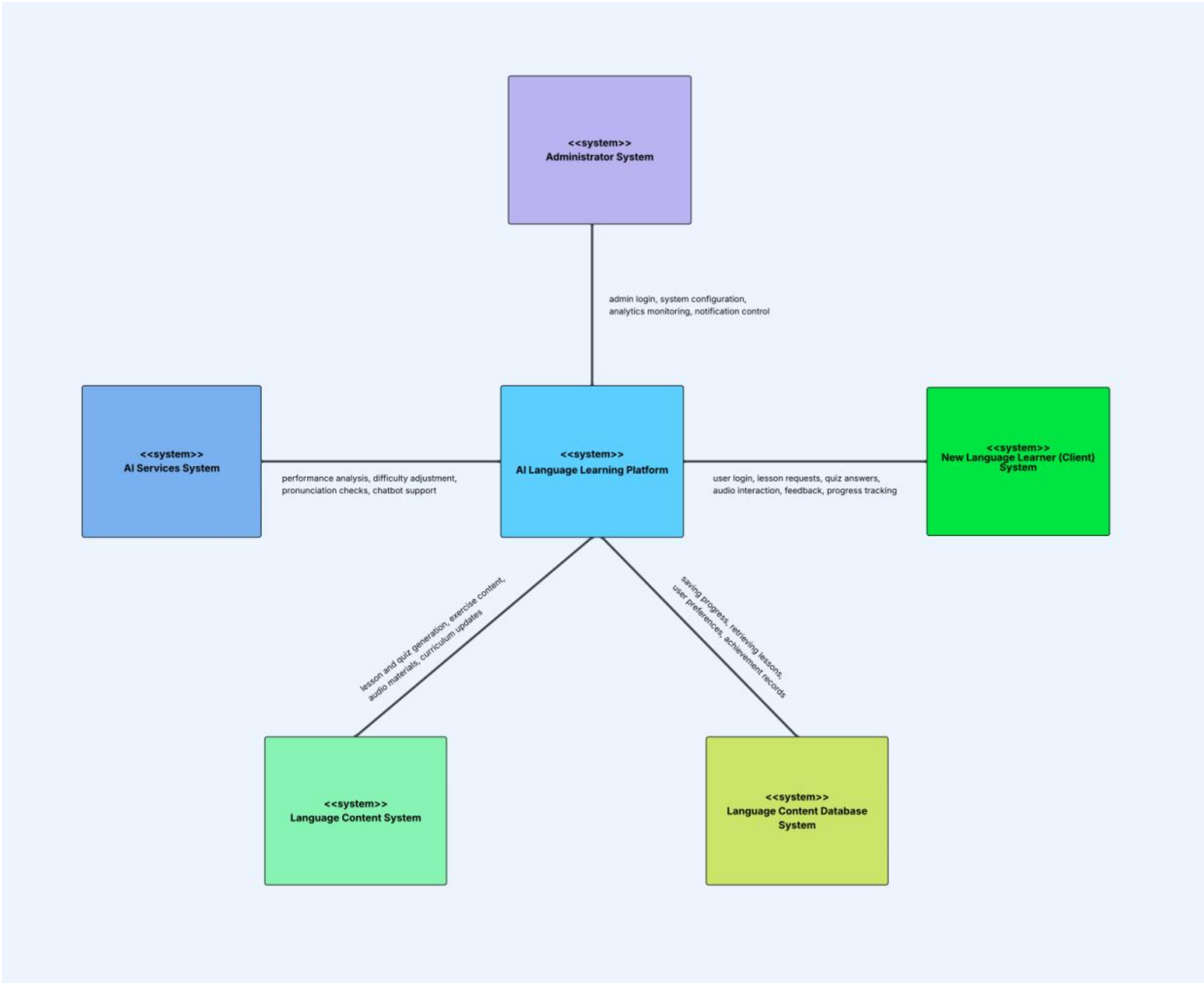
While market alternatives like Duolingo exist, our platform offers a more extensive feature set and greater global feasibility.

Our objective is to reach a global, multi-generational audience through effective teaching techniques. The novelty of our approach lies in the seamless integration of global languages. voice/audio capabilities, digital sketchpads, and AI-driven learning. From a technical standpoint, the product is built using a full-stack architecture- utilizing React for the frontend and Python for the backend- to ensure a highly interactive and user-friendly experience. This deep integration of AI distinguishes us from competitors, while individual client and admin login pages ensure that every user's progress is securely saved and personalized.

1.5 System Requirements

1.5.1 Context Diagram

Diagram 1:



Overview

This system follows a centralized platform model where the AI Language learning platform acts as the primary orchestrator. It sits at the heart of the architecture, providing communication between the user-facing interface, the administrative oversight tools, the backend content management, and the externalized AI services.

Key Components

Core Learning Platform

The central engine that connects all other modules.

- Key Functions: Handles user lifecycle(registration/login) manages the assessment logic for level placement, sequences lessons, and coordinates with the AI engine to adjust difficulty in real-time.

Administration & Monitoring

This is the control plane of the system. It interacts with the Core Platform to manage users and system configurations

- Key Functions: Admin authentication, system-wide configuration, and a dashboard for monitoring user analytics and sending platform-wide notifications.

AI Model & Services

This is the intelligence layer. This system processes raw data from the Core Platform to provide personalized learning experiences.

- Key Functions: Powers the recommendation engine, manages adaptive difficulty scaling, performs real-time pronunciation analysis, and hosts the chatbot for conversational practice.

Client Application (Learner Interface)

The front-end entry point for the end-user, built with modern web frameworks (React).

- Key Functions: Provides the UI for lessons and quizzes, handles audio input/output for speech practice, and visualizes progress through gamified elements like streaks and progress bars.

Content Management

The production factory for educational material. It sits between the AI services and the database.

- Key functions: Generates grammar and vocabulary exercises, manages audio clips for pronunciation, and ensures language-specific rules are applied to all generated content.

Learning Content Database

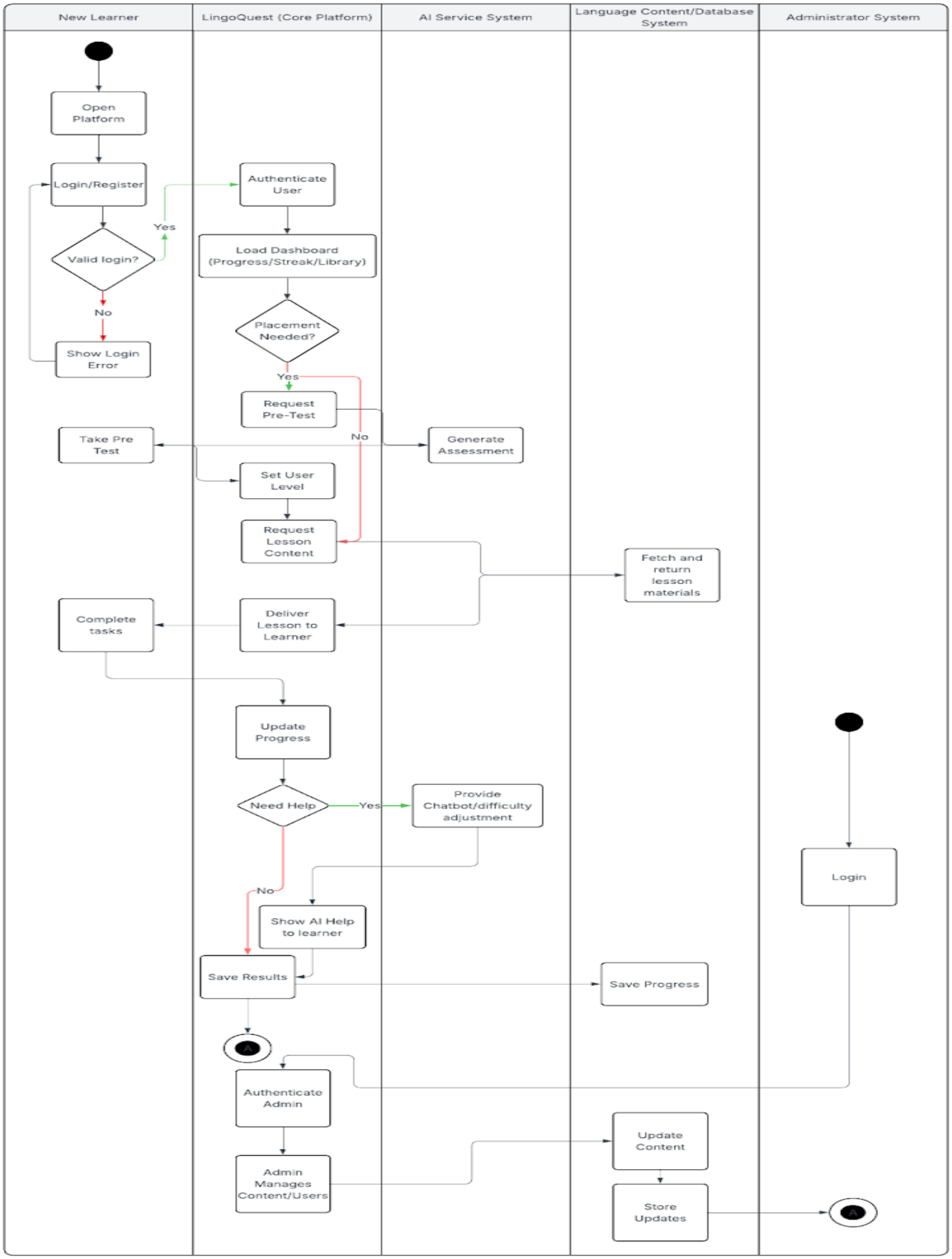
The persistence layer of the architecture. It serves as the single source of truth for both static and dynamic data.

- Key Functions: ZStores the lesson repository question banks, raw audio files, and critical user data, including progress records, preferences, and achievements.

Interaction Flow

1. The learner interacts with the Client Application, which requests content from the Core platform.
2. The Core platform fetches curriculum from Content Management and user history from the database.
3. Simultaneously, the AI Services analyze the learner's performance to suggest the next best step or adjust lesson difficulty.
4. The Administrator monitors the entire flow, updating content or system settings as needed through the admin System.

1.5.2 Activity Diagram


New Learner
LingoQuest (Core Platform)
AI Service System
Language Content/Database System
Administrator System
Open Platform
Login/Register
Valid login?
Yes
No
Show Login Error
Authenticate User
Load Dashboard (Progress/Streak/Library)
Placement Needed?
Yes
No
Request Pre-Test
Generate Assessment
Take Pre Test
Set User Level
Request Lesson Content
Fetch and return lesson materials
Deliver Lesson to Learner
Complete tasks
Update Progress
Need Help
Yes
Provide Chatbot/difficulty adjustment
No
Show AI Help to learner
Save Results
Save Progress
Login
Authenticate Admin
Admin Manages Content/Users
Update Content
Store Updates

2.0 REQUIREMENTS

2.1 Use Cases

1. **Use Case#007**

Name: User Management

Actors:

- New Language Learner (User) – creates account, logs in/out
- Core Learning Platform – validates credentials, creates sessions
- Learning Content Database – stores user profile, hashed password, progress, preferences

Use Case Description:

1. User opens the website and selects Register or Login.
2. If Register: user enters required info and submits.
3. System validates fields (format, password rules).
4. System stores the new user in the database and creates a profile.
5. If Login: user enters credentials and submits.
6. System verifies credentials and starts a secure session.
7. User can log out; the system ends the session and returns to the landing page.

Exception Path:

- If the email already exists, then show an error and suggest Login.
- Invalid password/username, then show error, allow retry.
- Server/database unavailable, then show "Try again later" and log the error.

Alternate Path:

- "Forgot password" flow (if implemented later), then send reset link.

Pre-condition:

- The website is reachable, and the database/service is available.

Post-condition:

- User is authenticated (or account is created) and can access learning features.

2. ***UseCase#020***

Name: Select Target Language (Start Learning Path)

Actors:

- Learner (User)
- Client Application (React UI)
- Content Management

- Learning Content Database

Use Case Description:

1. Logged-in user opens Language Selection.
2. User chooses a target language.
3. System loads learning content and initializes the user’s learning path for that language.
4. System confirms selection and redirects user to the pre-test or first lesson (depending on setup).

Exception Path:

- If the content for the language is not available, then show a message and suggest another language.

Alternate Path:

- User changes language later, then the system saves the current language progress and switches context.

Pre-condition:

- User is logged in.

Post-condition:

- The target language is set for the user profile

3.**Use Case#562**

Name: Take Skill Assessment Pre-Test (Placement)

Actors:

- Learner (User)
- Core Learning Platform
- AI Model & Services (recommendation/placement)
- Learning Content Database

Use Case Description:

1. User starts the Pre-Test.
2. The system displays a series of skill questions for the chosen language.
3. User submits answers.
4. System scores the test and sends results to AI services.
5. AI recommends a starting level (beginner/intermediate/etc).
6. The system stores placement and redirects the user to the recommended starting lesson.

Exception Path:

- User exits mid-test, then saves partial progress or marks as incomplete.
- AI service fails, so use rule-based fallback placement (basic scoring bands).

Alternate Path:

- User skips pre-test, then defaults to beginner level.

Pre-condition:

- User is logged in and has selected a target language.

Post-condition:

- The starting level is assigned and saved.

4.**Use Case #24**

Name: Complete a Lesson (Level-Based Learning)

Actors:

- Learner (User)
- Client Application
- Core Learning Platform
- Content Management
- Learning Content Database

Use Case Description:

1. User selects a lesson from their assigned level.
2. System loads lesson content (vocab/grammar).
3. User reviews lesson pages and examples.
4. The user completes the lesson and proceeds to the quiz/checkpoint.
5. System saves lesson completion status.

Exception Path:

- Content fails to load, then shows retry and log error.

Alternate Path:

- User revisits completed lesson, then system loads in "review mode."

Pre-condition:

- User has an assigned level (from pre-test or default).

Post-condition:

- Lesson marked completed; progress updated.

5. **Use Case #645**

Name: Adaptive Quiz (AI Adjusts Difficulty)

Actors:

- Learner (User)
- Core Learning Platform (quiz engine)
- AI Model & Services (adaptive difficulty)
- Learning Content Database

Use Case Description:

1. User begins a quiz for the current lesson/level.
2. The system presents questions.
3. User submits answers.
4. The system calculates accuracy and sends performance to AI services.
5. AI adjusts future question difficulty based on accuracy (easier/harder).
6. System saves quiz result and updates mastery/progress.

Exception Path:

- User submits an invalid/empty answer, then prompt user to answer before continuing.
- AI service unavailable, then continue withthe standard difficulty set.

Alternate Path:

- If the user scores above the threshold, then unlock the next level/unit.
- If the user scores below the threshold, then recommend a review lesson.

Pre-condition:

- Lesson content exists, and the user is in an active learning path.

Post-condition:

- Quiz results stored; difficulty profile updated.

6.**Use Case #966**

Name: Use Sketchpad for Lesson Notes

Actors:

- Learner (User)
- Client Application (Sketchpad UI)
- Learning Content Database (note persistence)

Use Case Description:

1. User opens sketchpad from lesson screen.
2. User writes/draws notes during the lesson.
3. User selects Save.
4. The system stores sketchpad content linked to the lesson and the user account.
5. The user can reopen saved notes later.

Exception Path:

- Save fails, then show error and keep notes in local session until retry.

Alternate Path:

- User chooses “Clear” then the sketchpad resets after confirmation.

Pre-condition:

- User is logged in and viewing a lesson.

Post-condition:

- Notes saved and tied to that lesson/unit.
-

7. Use Case #17

Name: Pronunciation Test with Audio

Actors:

- Learner (User)
- Client Application (mic/audio UI)
- AI Model & Services (pronunciation analysis)
- Learning Content Database (store attempts/results)

Use Case Description:

1. User starts a pronunciation exercise.
2. System requests microphone permission.
3. User records pronunciation and submits.
4. The system sends audio to an AI service for analysis.
5. AI returns a score/feedback (accuracy, suggested corrections).
6. System displays feedback and stores attempt history.

Exception Path:

- Mic denied, then provide fallback (listen-only mode or typed practice).
- Audio upload fails, then prompt retry.

Alternate Path:

- User reattempts pronunciation, then keep best score or average (config-based).

Pre-condition:

- User is logged in; device supports audio input.

Post-condition:

- Attempt stored; progress may update if threshold met.

8.**Use Case#501**

Name: Track Progress (Streaks, Progress Bars, Milestones)

Actors:

- Learner (User)
- Client Application (progress UI)
- Core Learning Platform
- Learning Content Database

Use Case Description:

1. User completes a lesson/quiz.
2. System updates completion counters and mastery status.
3. System updates streak based on daily activity rules.
4. System updates progress bars for level/unit completion.
5. If a milestone is reached, the system generates an award/share prompt and plays a feedback sound.

Exception Path:

- Progress update fails, then queue update and retry; show minimal progress until sync.

Alternate Path:

- Streak breaks (missed day), then resetthe streak and show an encouragement message.

Pre-condition:

- User has performed a learning activity (lesson/quiz).

Post-condition:

- Progress and streak values are stored and displayed.

9.**Use Case #102**

Name: Admin Monitoring & Platform Notification Broadcast

Actors:

- Administrator
- Administration & Monitoring System (Admin Dashboard)
- Core Learning Platform
- Learning Content Database
- Email/Notification Service

Use Case Description:

1. Admin logs into the admin portal.
2. Admin opens dashboard to view user analytics (activity, progress trends).
3. Admin selects "Send Notification" (reminder/newsletter/system update).
4. Admin writes a message and selects the audience (all users / specific segment).
5. System validates content and queues delivery.
6. Notification is sent and logged for auditing.

Exception Path:

- Invalid admin credentials, then deny access.
- The notification service fails, then queue message and show failure status.

Alternate Path:

- Admin sends only in-app notifications (no email) based on user preferences.

Pre-condition:

- Admin account exists, and admin is authenticated.

Post-condition:

- Analytics viewed; notification broadcast recorded and delivered/queued.

10.**Use Case#85**

Name: Chatbot Conversation Help (Tasks & Language Q/A)

Actors:

- Learner (User)
- AI Chatbot Service
- Core Learning Platform

Use Case Description:

1. User opens chatbot panel.
2. User asks a question (translation, grammar, what to do next, general help).
3. System sends message + context to AI chatbot.
4. Chatbot responds with guidance.
5. User continues conversation or closes chat.

Exception Path:

- AI service error then shows "Chatbot unavailable" and suggest help page/FAQ.

Alternate Path:

- User asks for lesson-specific help, then chatbot links to lesson or recommends review.

Pre-condition:

- User is logged in; chatbot service available.

Post-condition:

- User receives guidance; optionally store conversation history.

2.2 Requirements

Requirement number: 1
Use Case#007

Introduction: User Management enables users to securely create accounts, log in, maintain their personal progress, and safely log out. This requirement ensures secure access control.

Inputs:

- Registration: name/username, email, password
- Login: email/username, password
- Logout: active session/token

Requirements Description:

The system allows users to register using an email address, username, and password. During registration, the system will validate email format and password requirements. User passwords will be securely hashed before being stored in the database to protect the information. The system will authenticate users using their valid credentials and establish a secure login session. Users will be able to log out at any time, which will terminate the session.

Outputs:

- Account created OR login session established
- User dashboard loads with saved progress/preferences

Requirement number: 2
Use Case#020

Introduction: The platform must allow users to select a language to start their learning experience and access language-specific lessons.

Inputs:

- Selected language (e.g., Spanish, French)

Requirements Description:

The system allows users to register using an email address, username, and password. During registration, the system will validate the email format and password requirements. User passwords will be securely hashed before being stored in the database to protect the information. The system will authenticate users using their valid credentials and establish a secure login session. Users will be able to log out at any time, which will terminate the session.

Outputs:

- User sees language-specific dashboard/starting screen

Requirement number: 3
UseCase#562

Introduction: The pre-test evaluates a learner's proficiency level and assigns the appropriate starting point within the curriculum to ensure personalized and effective instruction.

Inputs:

- User answers to pre-test questions

Requirement Description:

The system will present a set of assessment questions designed to evaluate the learner's knowledge of the selected language. After submission, the system will calculate a score based on the user's responses and assign a proficiency level such as Beginner, Intermediate, or Advanced. Users have the option to skip the pre-test and default to the Beginner level. Placement results will be generated accurately and within a reasonable response time.

Outputs:

- Display recommended starting level and next-step lesson link

Requirement number: 4
UseCase#24

Introduction: Lessons are organized into structured levels and units to guide learners progressively through the curriculum.

Inputs:

- Lesson selection (level/unit)
- User interactions (next, review, complete)

Requirement Description

The system will display lessons according to the user's assigned proficiency level and allow navigation through lesson pages in a logical sequence. Upon completion of a lesson, the system will mark it as complete and store this status in the database. Users will be able to revisit previously completed lessons for review at any time.

Outputs:

- Updated progress bar/lesson status saved

Requirement number:5
UseCase#645

Introduction: The adaptive quiz system personalizes learning by adjusting question difficulty based on user performance.

Inputs:

- User quiz answers
- Accuracy score/response history

Requirement Description

After completing a lesson, the system will present quiz questions related to the lesson content. The system will evaluate user responses, calculate accuracy, and adjust the difficulty of subsequent questions based on performance results. If a user achieves a defined performance threshold, the system will unlock higher-level content. If performance falls below expectations, the system will recommend reviewing lesson material before progressing.

Outputs:

- Score + feedback shown; next lesson recommendation updated

Requirement number:6
UseCase#966

Introduction: The sketchpad feature enhances engagement by allowing learners to take notes or practice writing directly within lessons.

Inputs:

- Sketch strokes/typed notes
- Save /clear/export action

Requirement Description

The system will provide a digital canvas that supports writing or drawing using mouse input. Users will be able to create, edit, and clear notes during lessons. The system will allow users to save notes and associate them with specific lessons for later retrieval. Saved sketchpad content will be persistently stored in the database and remain accessible whenever the user revisits the corresponding lesson.

Outputs:

- Saved sketchpad notes are available in the lesson review

Requirement number: 7
UseCase#17

Introduction: The pronunciation feature enables learners to practice speaking skills and receive AI-generated feedback.

Inputs:

- Microphone permission
- User recorded audio sample

Requirement Description

The system will request permission to access the user's device microphone and allow audio recording during pronunciation exercises. Recorded audio will be securely transmitted to an AI evaluation service for analysis. The system will generate a pronunciation score and provide corrective feedback based on the analysis.

Outputs:

- Pronunciation score + feedback + stored attempt record

Requirement number: 8
UseCase#501

Introduction: The progress tracking system motivates learners by visually representing achievements, milestones, and consistency through streaks.

Inputs:

- Completed lessons/quizzes
- Daily activity timestamps

Requirement Description

The system will update user progress automatically upon completion of lessons and quizzes. It will calculate and maintain daily streak counts based on user activity timestamps. The system will display progress bars, achievement badges, and milestone notifications to encourage continued engagement.

Outputs:

- Updated streak count, progress bar, milestone award/feedback

Requirement number: 9
UseCase#102

Introduction: The admin monitoring system allows administrators to oversee platform activity and communicate with users when necessary.

Inputs:

- Admin login credentials
- Notification content (title, message, target group)
- Analytics filters (date range, language, level)

Requirement Description

The system will provide administrators with secure, role-based access to an administrative dashboard. The dashboard will display analytics such as user activity levels and lesson completion rates. Administrators will be able to send in-app notifications and email messages to all users or selected user groups. Access to the administrative system will be restricted to authorized personnel only, ensuring platform security and integrity.

Outputs:

- Dashboard metrics displayed + notification sent/logged

Requirement number: 10
Use Case#85

Introduction: The AI chatbot provides real-time conversational assistance and learning support within the platform.

Inputs:

- User message/question
- Conversation context (current lesson/task)

Requirement Description

The system will include a chat interface that allows users to submit questions related to grammar, vocabulary, or lesson instructions. The chatbot will generate AI-based responses relevant to the lesson context and maintain conversation continuity during an active session. The system will implement input and output filtering to prevent unsafe or inappropriate responses.

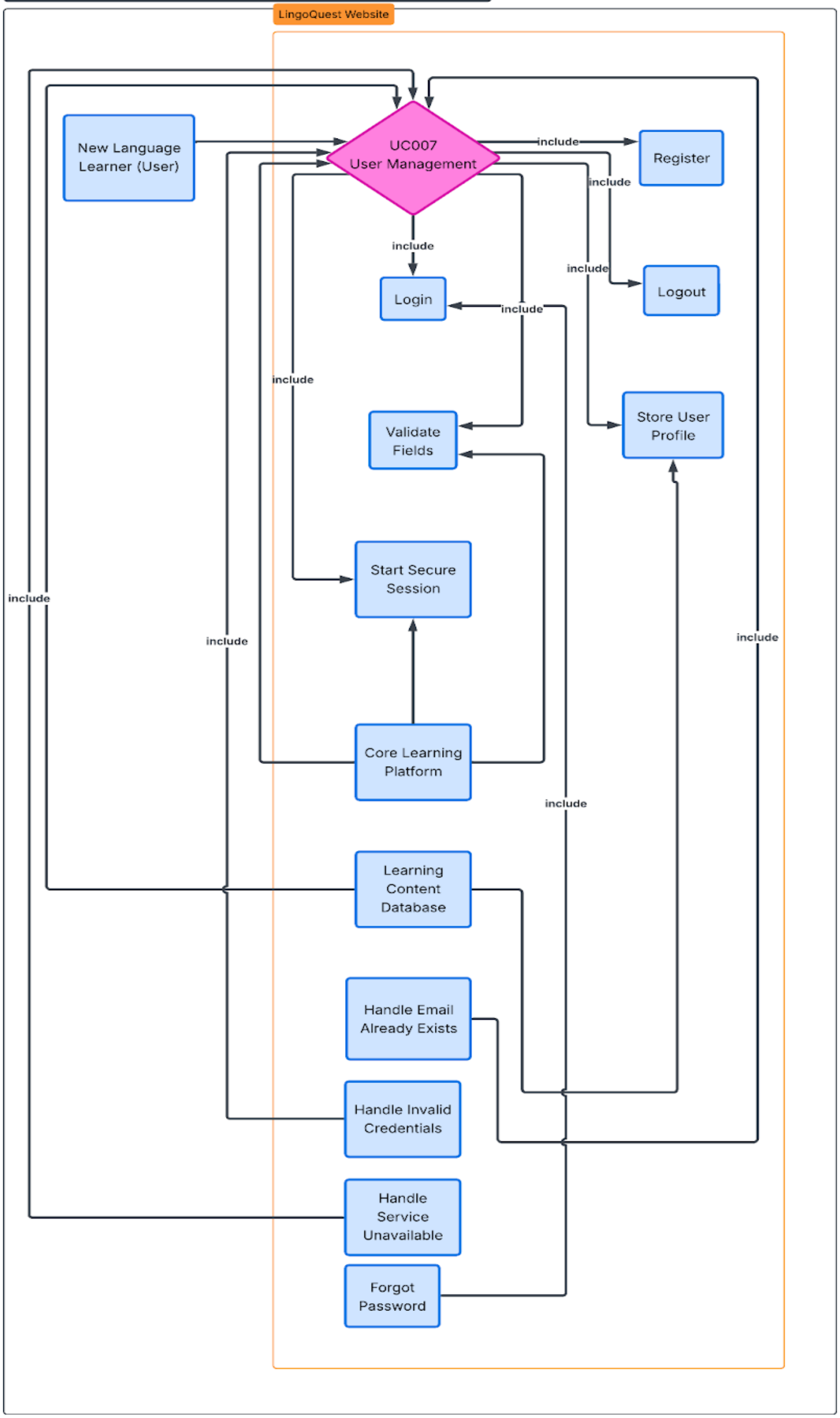
Outputs:

- Chatbot response displayed in UI

2.3 Use Case Diagrams

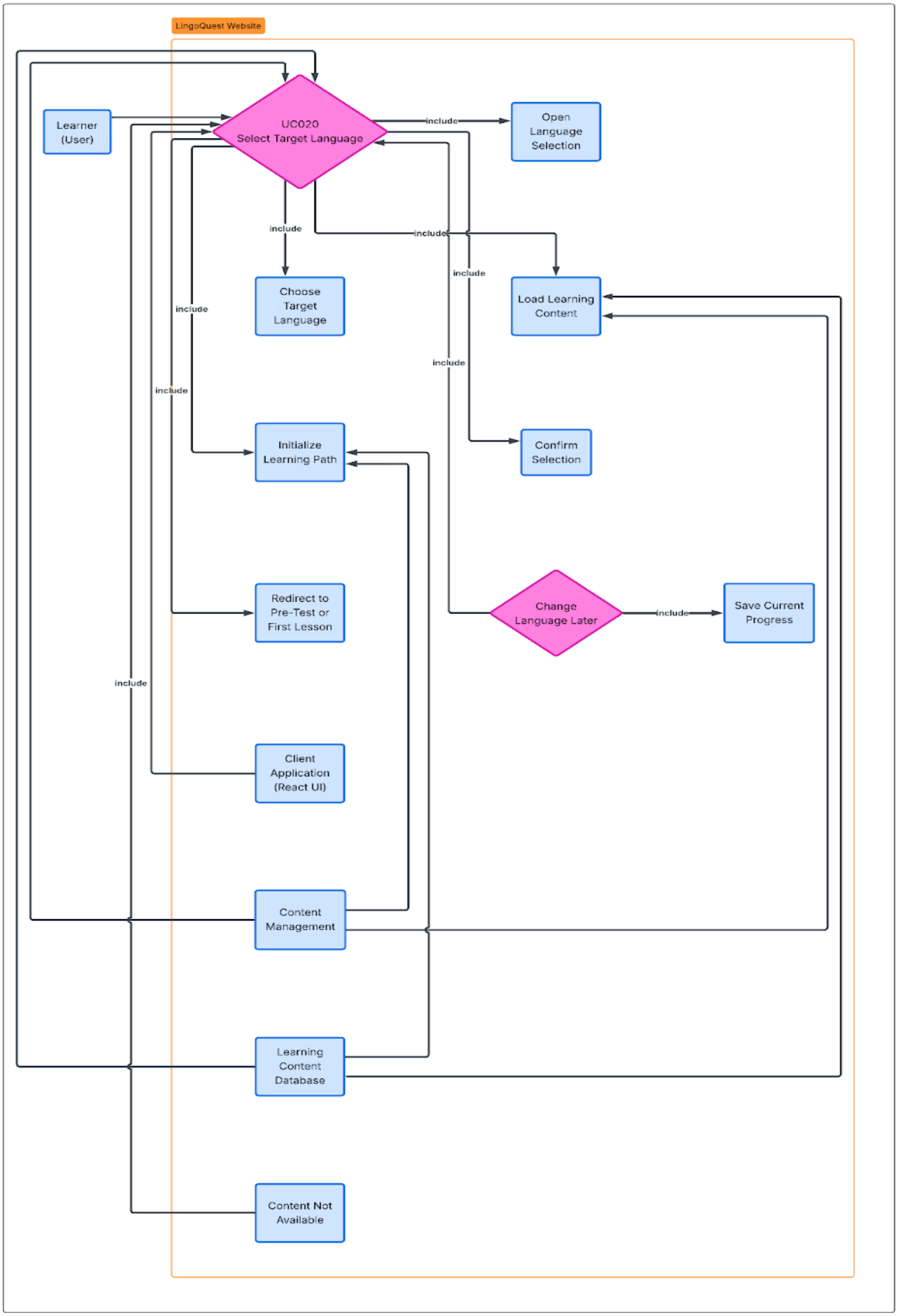
UseCase#007

User Management Use Case Diagram



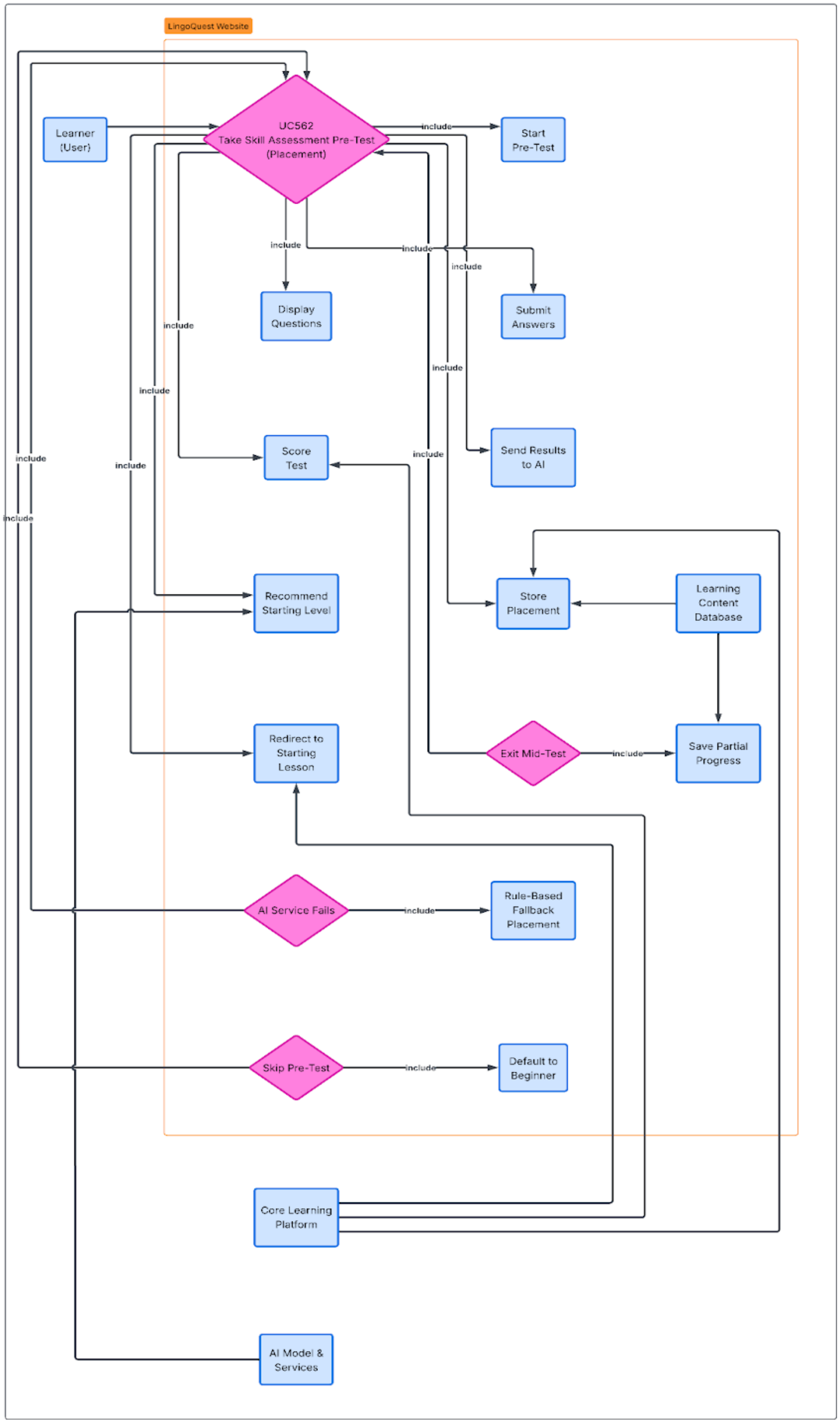
UseCase#020

Select Target Language Use Case Diagram



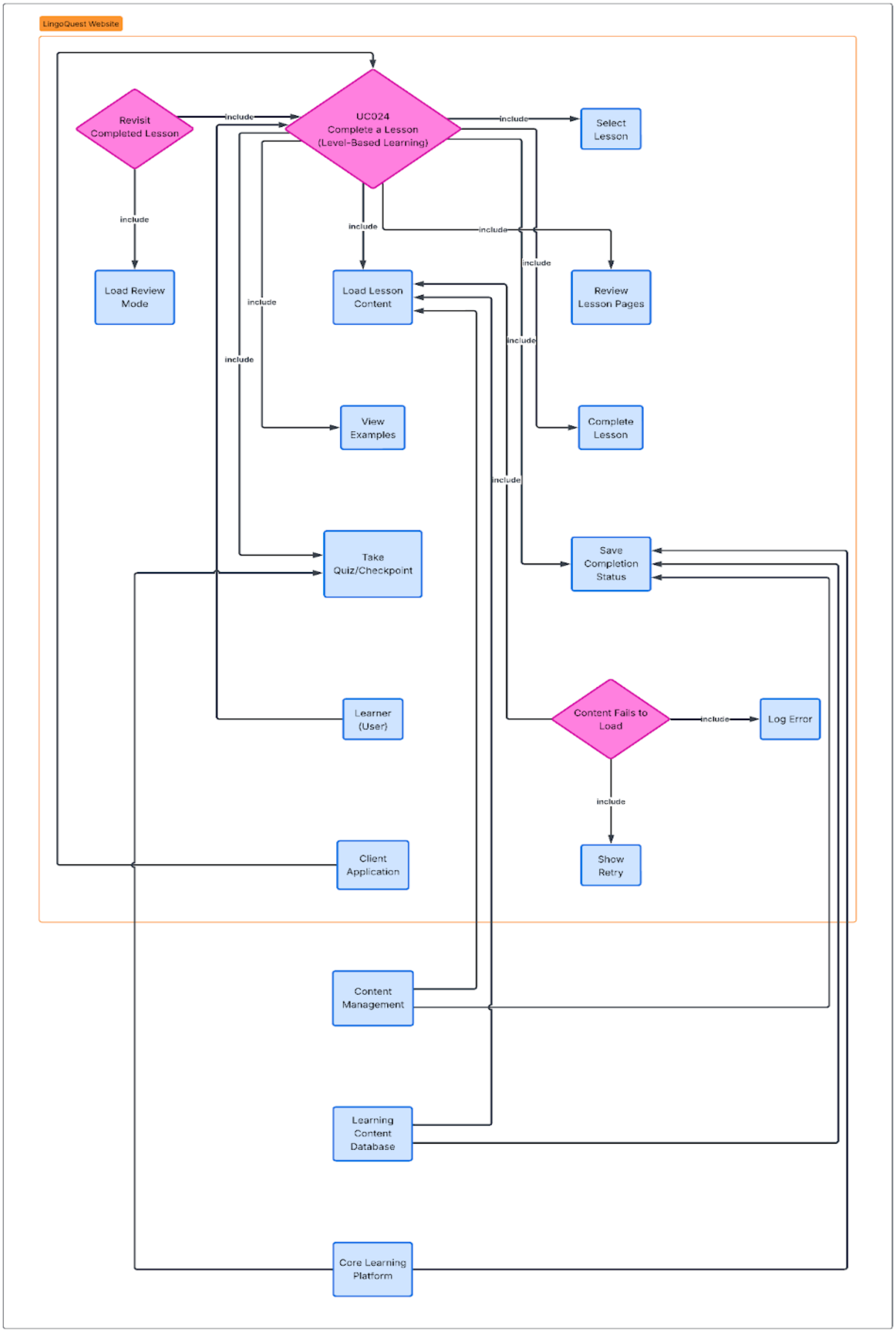
UseCase#562

Skill Assessment Pre-Test Use Case Diagram



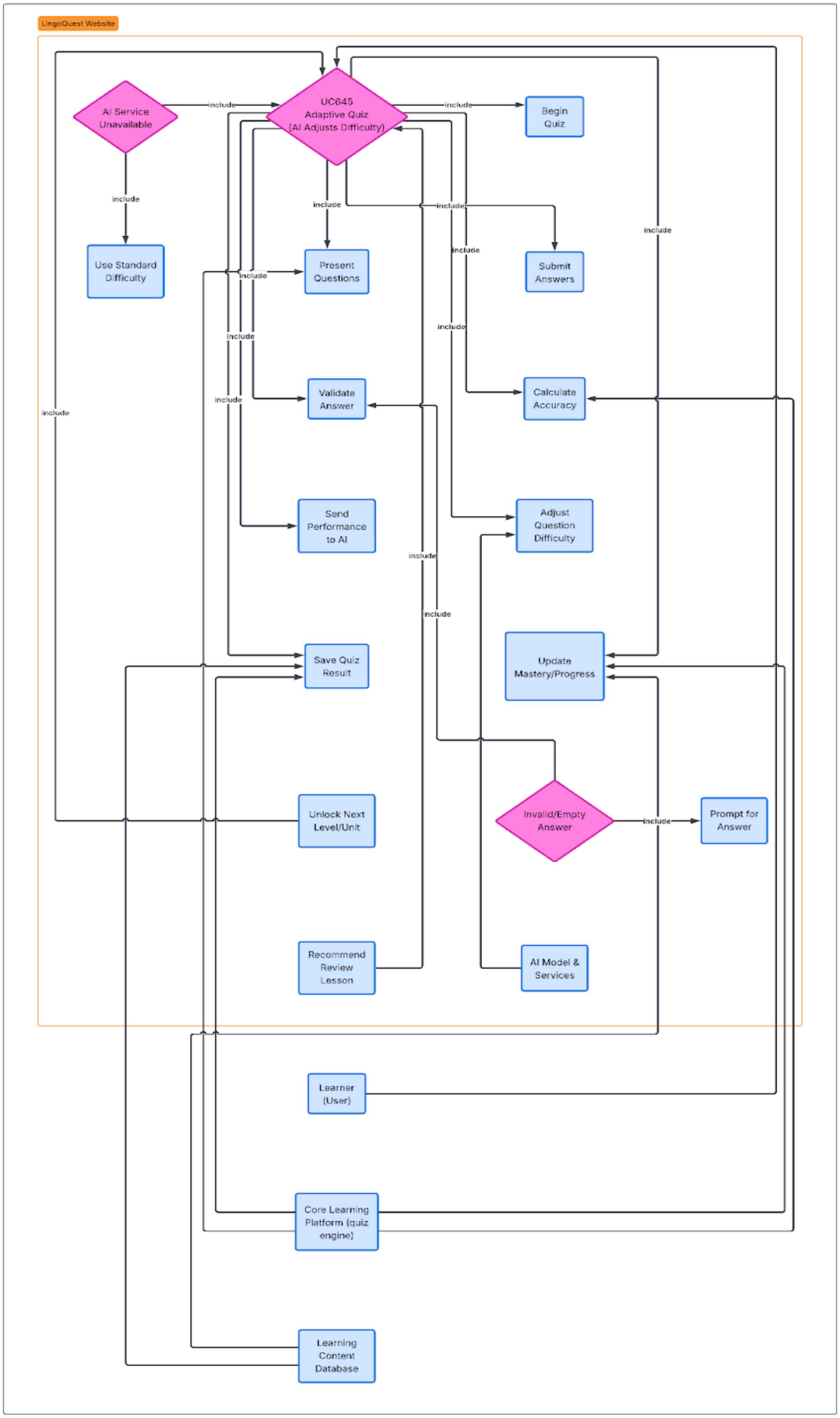
UseCase#24

Complete Lesson Use Case Diagram



UseCase#645

Adaptive Quiz Use Case Diagram



2.4 Database Management

- Type of database management system:
  - For this project, we will use a relational database implemented in MySQL. MySQL is widely supported and integrates well with our Python backend and React frontend.
- System Database Tables
  - Users table

```
CREATE TABLE IF NOT EXISTS users (
        user_id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
        first_name VARCHAR(50) NOT NULL,
        last_name VARCHAR(50) NOT NULL,
        email VARCHAR(100) NOT NULL UNIQUE,
        password_hash TEXT NOT NULL,
        role TEXT NOT NULL DEFAUlLT 'CLIENT',
        created_at TIMESTAMP NOT NULL DEFAULT NOW(),
        last_login TIMESTAMP NULL
);
```

  - Learning Paths

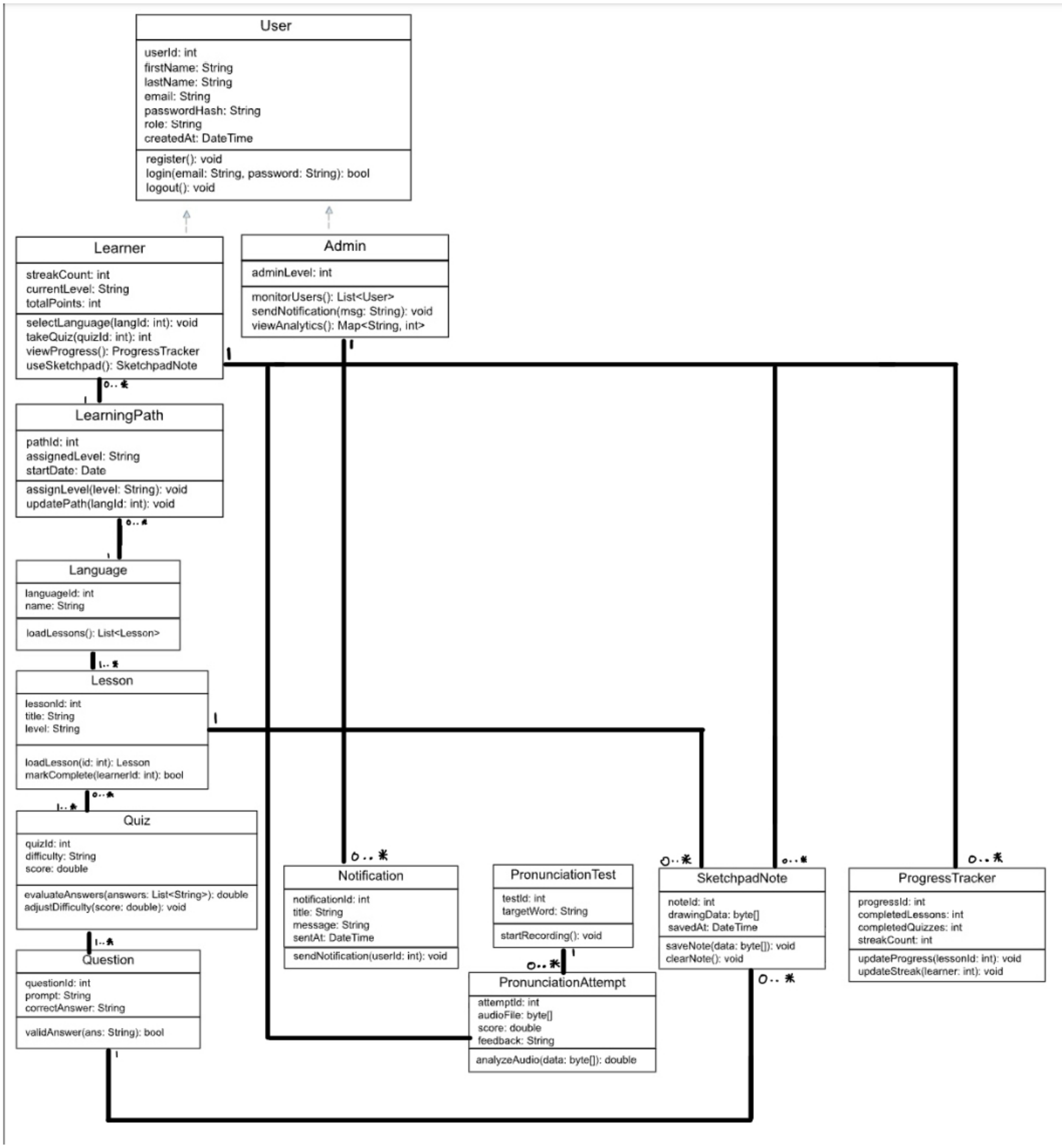
```
CREATE TABLE IF NOT EXISTS learning_paths (
        path_id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
        user_id BIGINT NOT NULL, assigned_level VARCHAR(50) DEFAULT 'Beginner',
        start_date TIMESTAMP NOT NULL DEFAULT NOW(),
        CONSTRAINT fk_learning_paths_user
                FOREIGN KEY (user_id)
                REFERENCES users(user_id)
                ON DELETE CASCADE
);
```

  - Progress tracker

```
CREATE TABLE IF NOT EXISTS progress_tracker (
        progress_id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
        user_id BIGINT NOT NULL,
        completed_lessons INT NOT NULL DEFAULT 0,
        completed_quizzes INT NOT NULL DEFAULT 0,
        streak_count INT NOT NULL DEFAULT 0,
        CONSTRAINT fk_progress_tracker_user
                FOREIGN KEY (user_id)
                REFERENCES users(user_id)
                ON DELETE CASCADE
);
```

Role: These are the base tables for authentication and role-based access control. Several other tables can be referenced through the user_id.

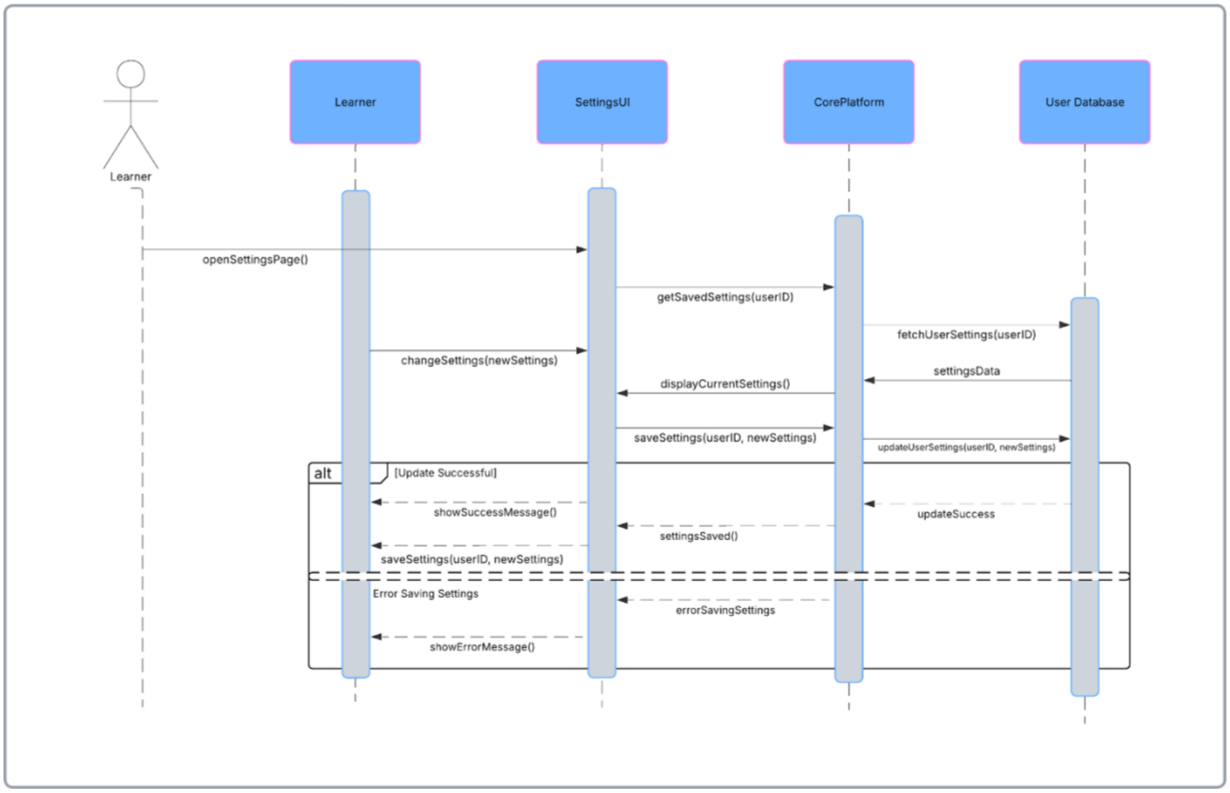
3.1 Class Diagrams



3.2 Behavior Modeling

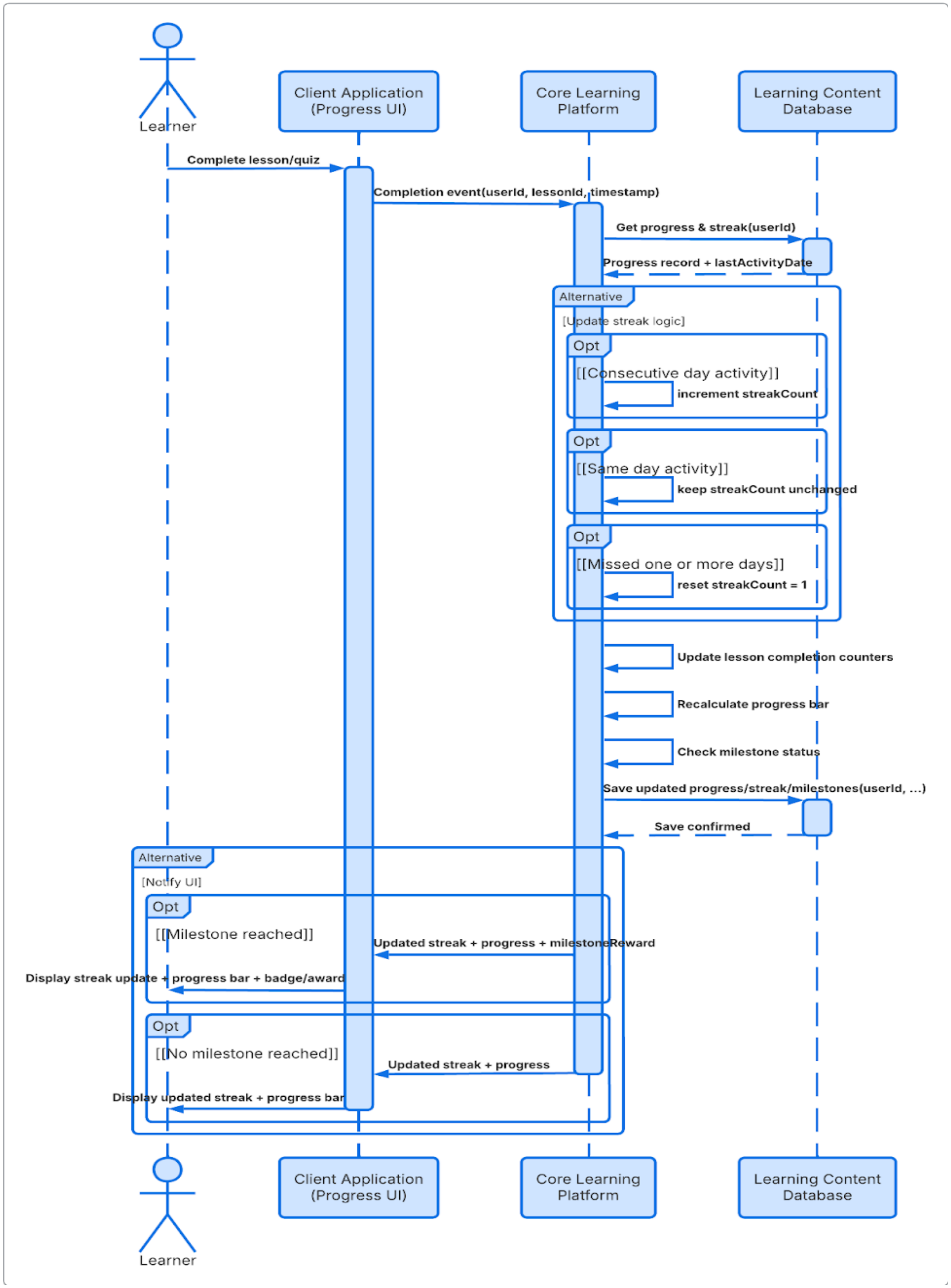
Settings Page UML Sequence Diagram (Faysal)

Sequence diagram



Daily Streaks Sequence Diagram (Harry)

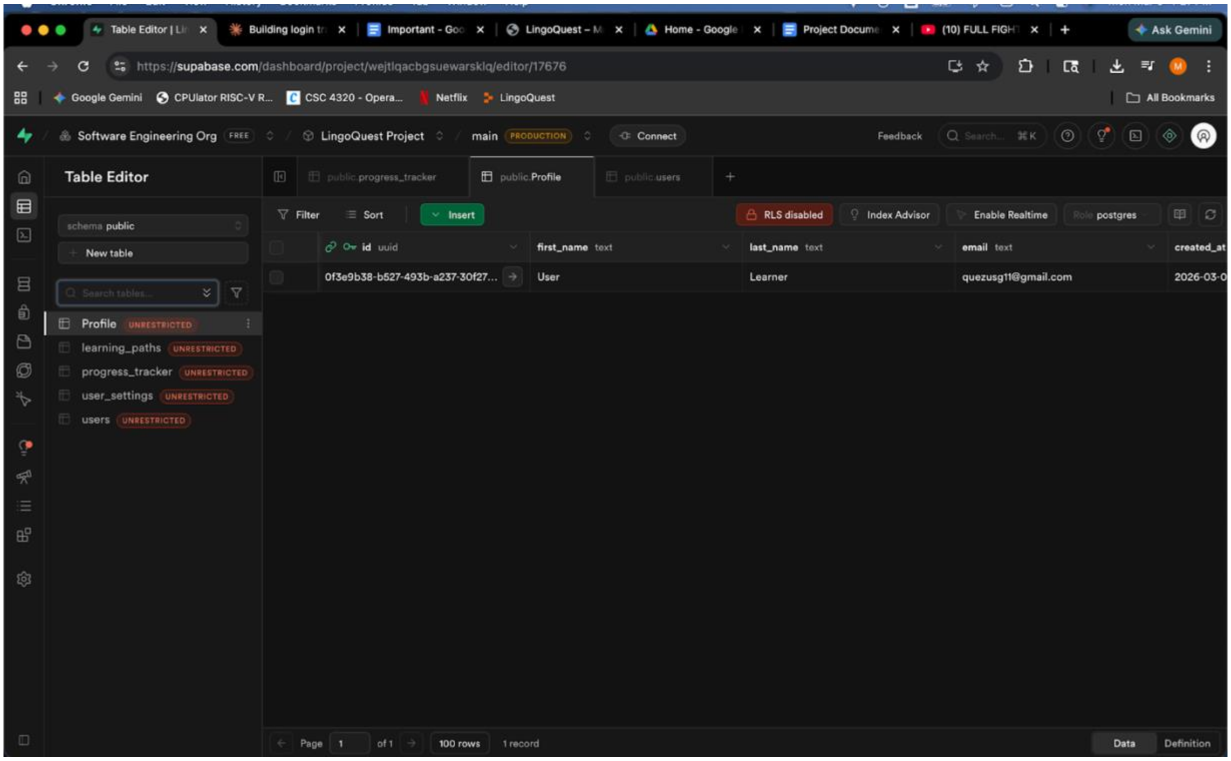
Daily Streaks Progress Tracking



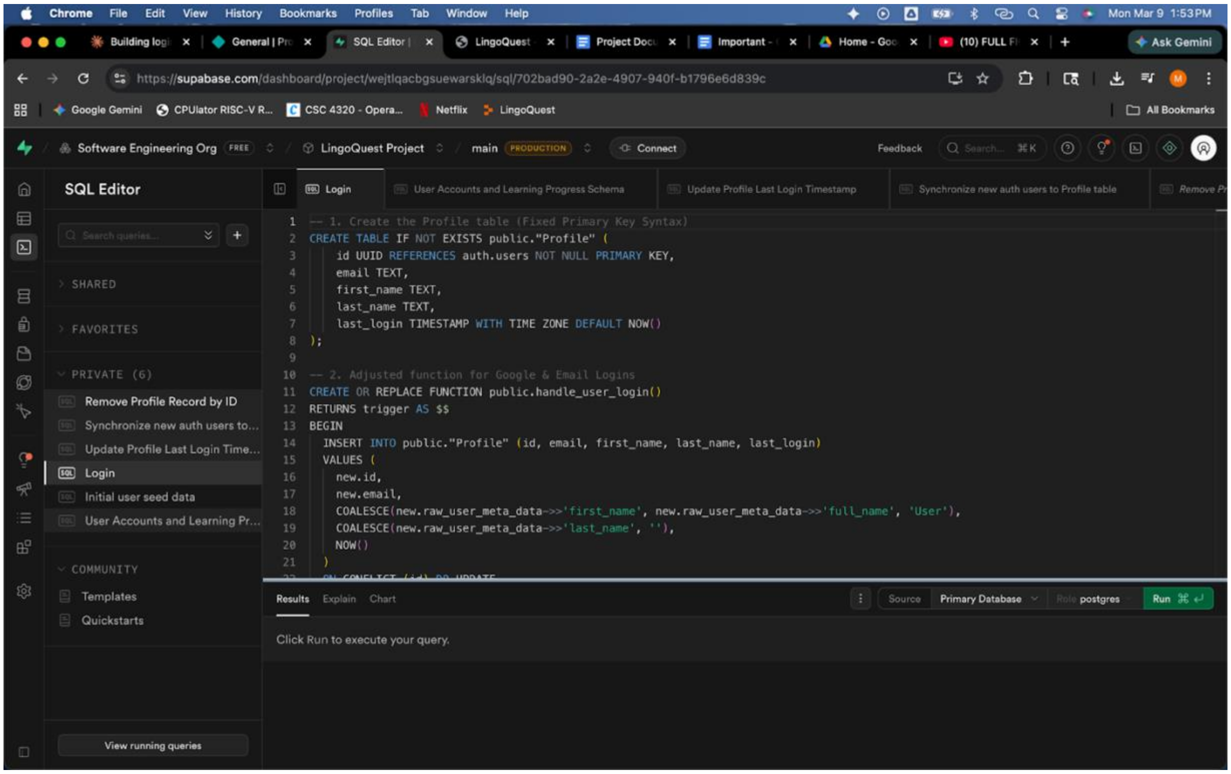
4.0 Testing

4.1 Implementation

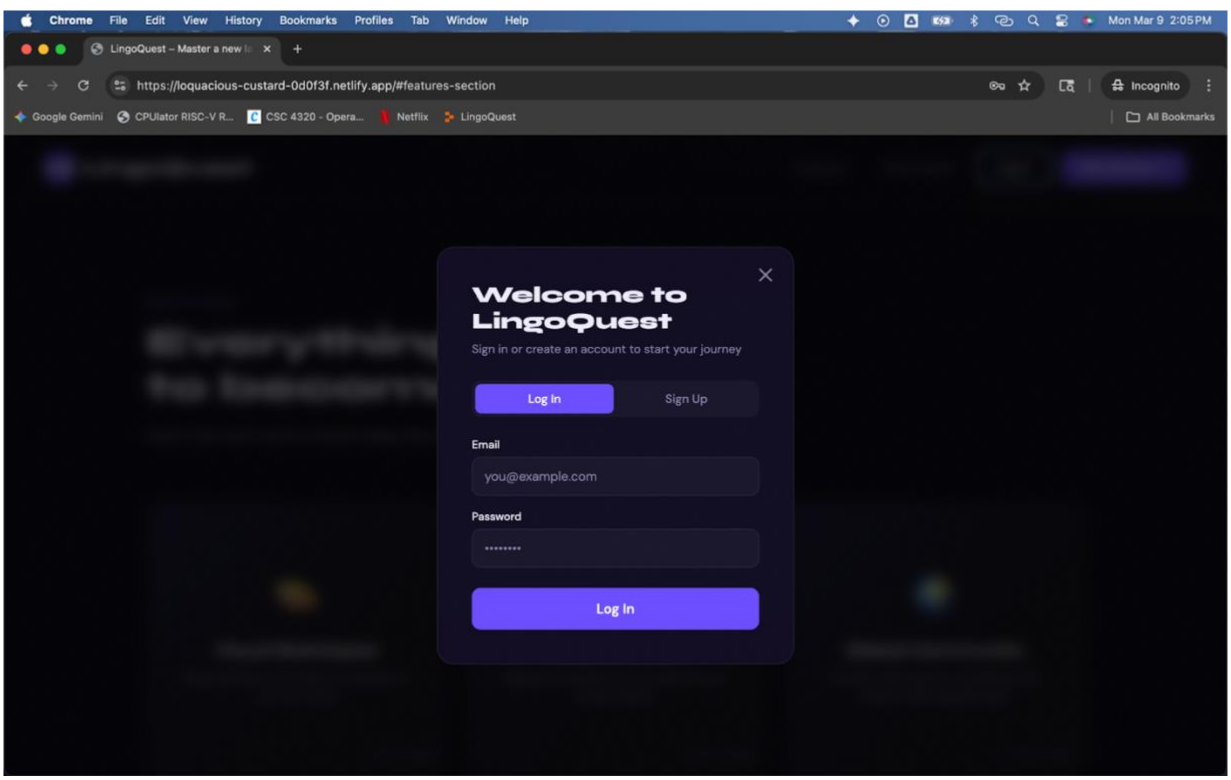
Proof of life:



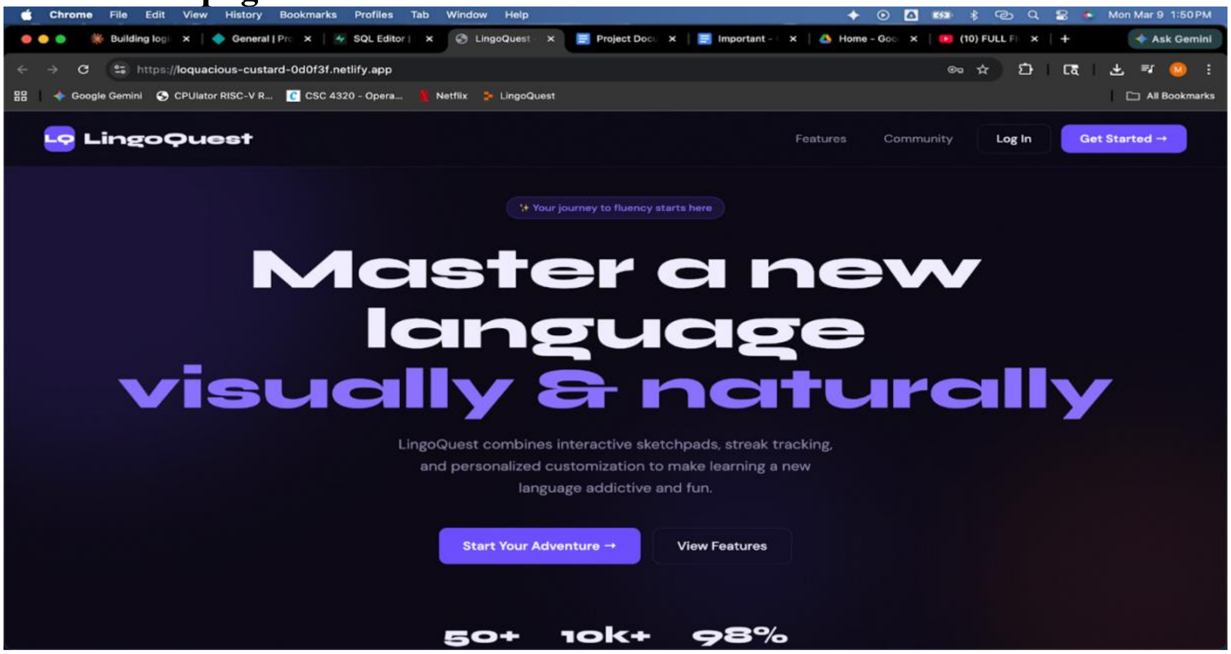
MySQL:



Front-end registration:

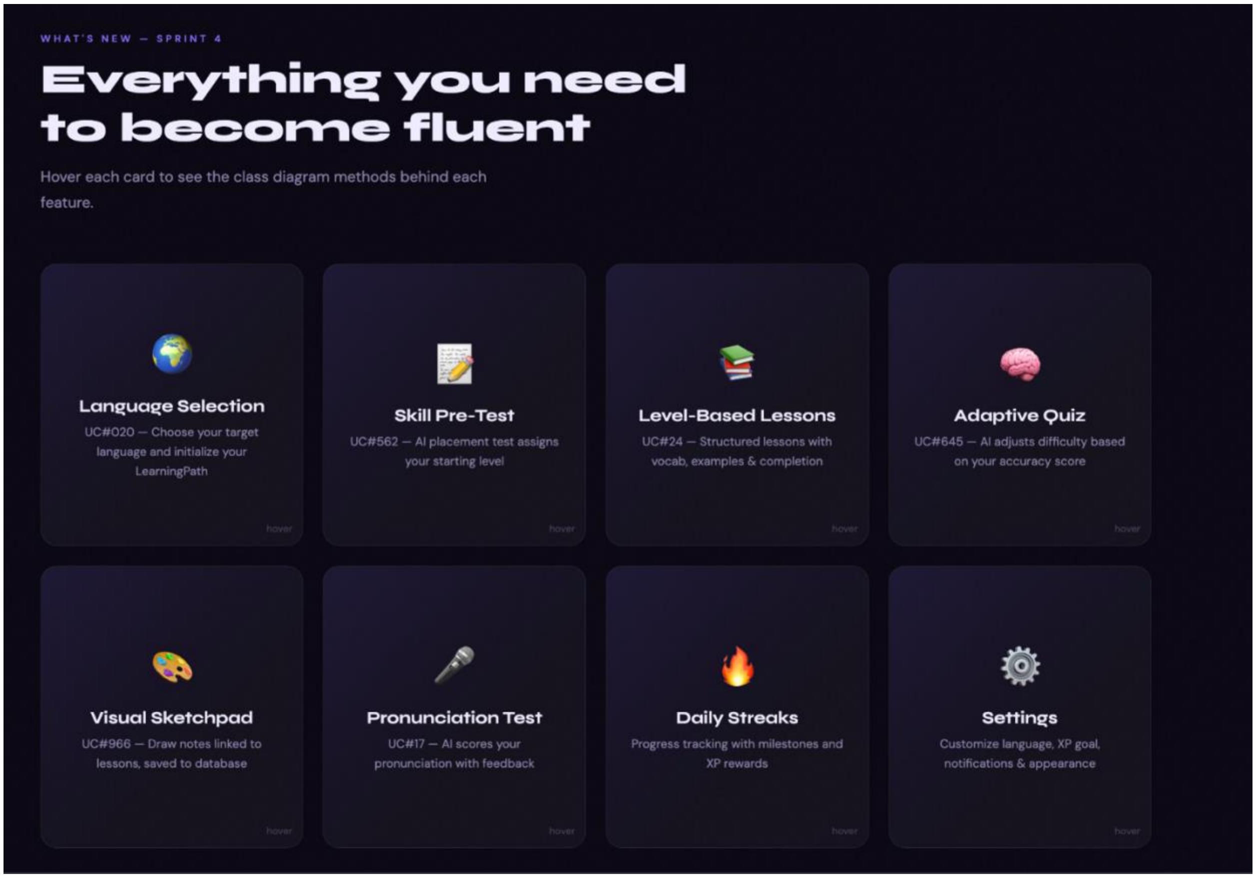


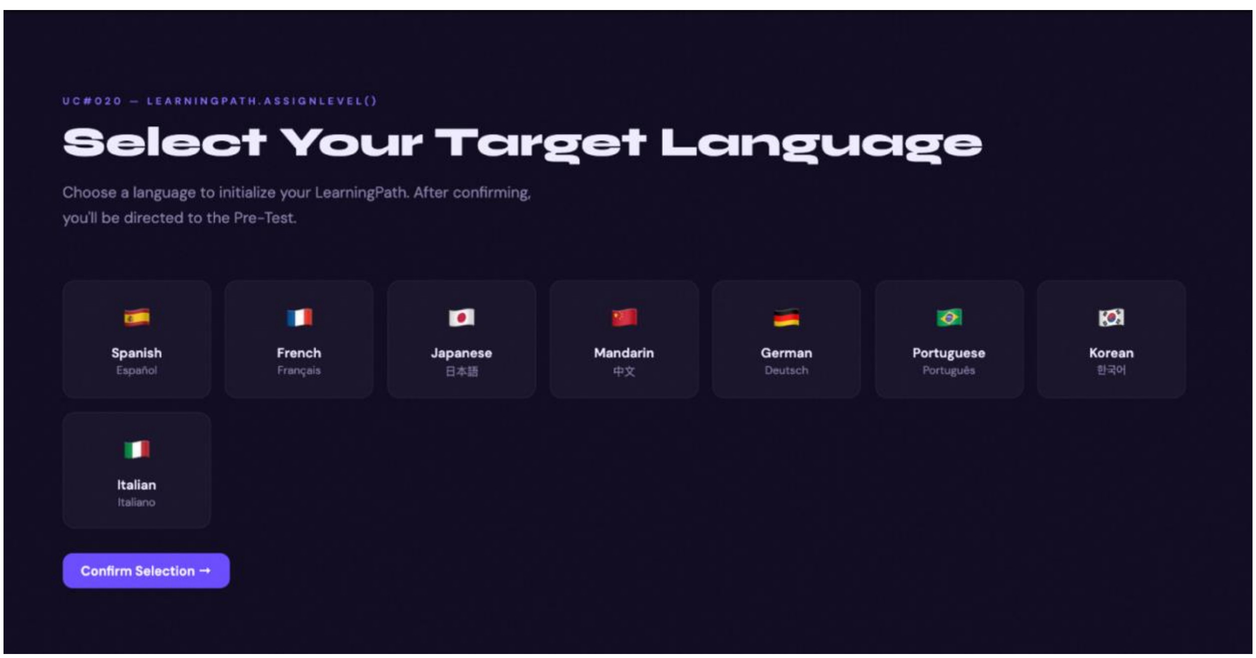
Front-end Home page:



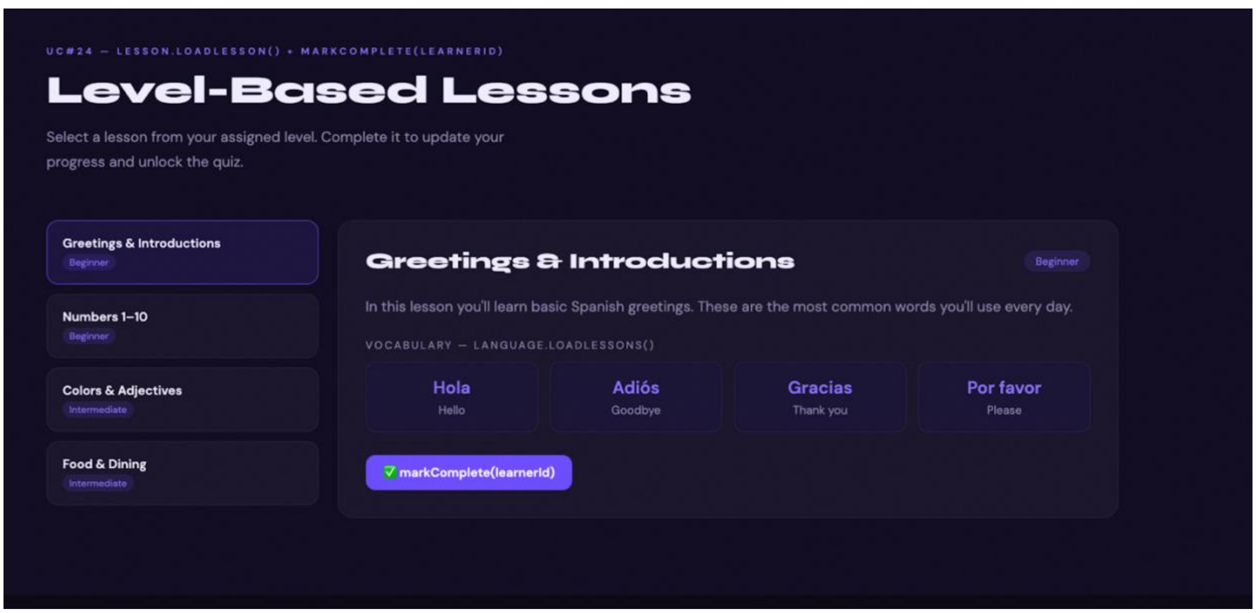
Features(Daily Streaks, Sketchpad, Global Community, Etc)

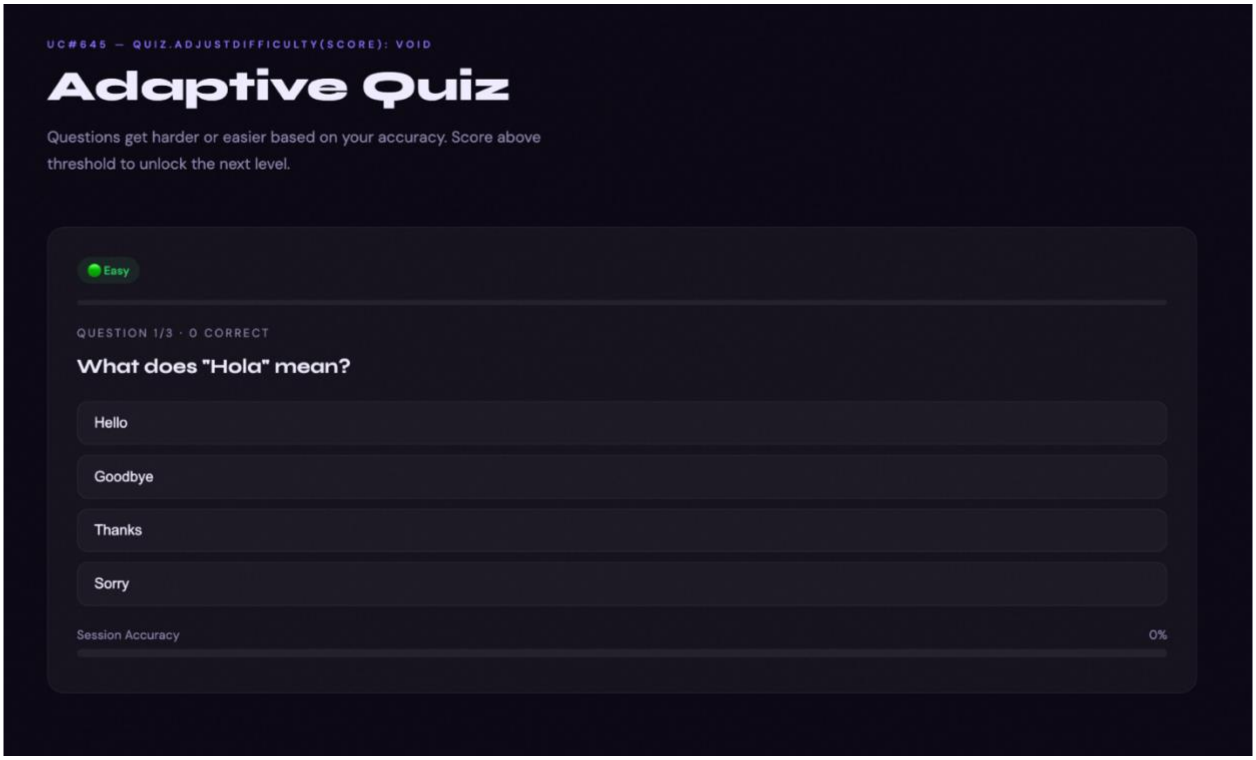
Settings:

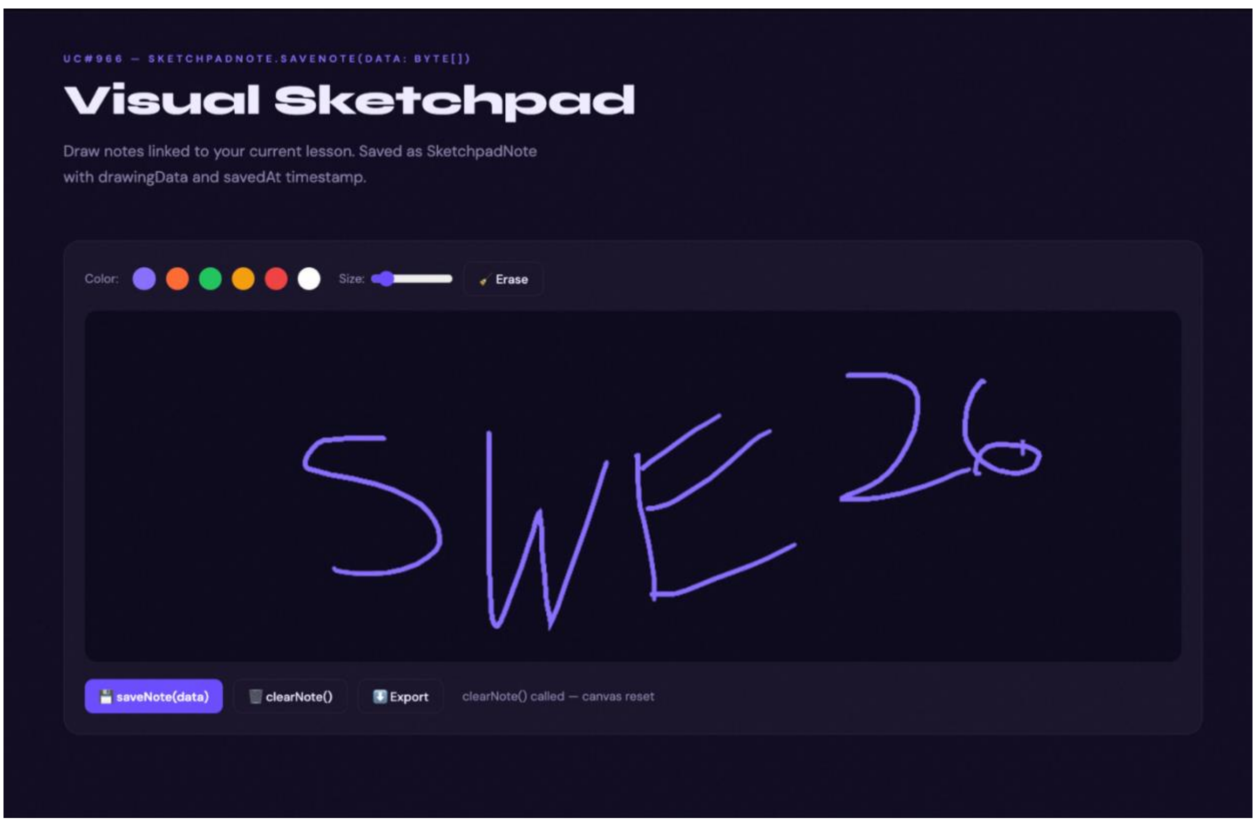

WHAT'S NEW — SPRINT 4
Everything you need to become fluent
Hover each card to see the class diagram methods behind each feature.
Language Selection
UC#020 — Choose your target language and initialize your LearningPath
hover
Skill Pre-Test
UC#562 — AI placement test assigns your starting level
hover
Level-Based Lessons
UC#24 — Structured lessons with vocab, examples & completion
hover
Adaptive Quiz
UC#645 — AI adjusts difficulty based on your accuracy score
hover
Visual Sketchpad
UC#966 — Draw notes linked to lessons, saved to database
hover
Pronunciation Test
UC#17 — AI scores your pronunciation with feedback
hover
Daily Streaks
Progress tracking with milestones and XP rewards
hover
Settings
Customize language, XP goal, notifications & appearance
hover

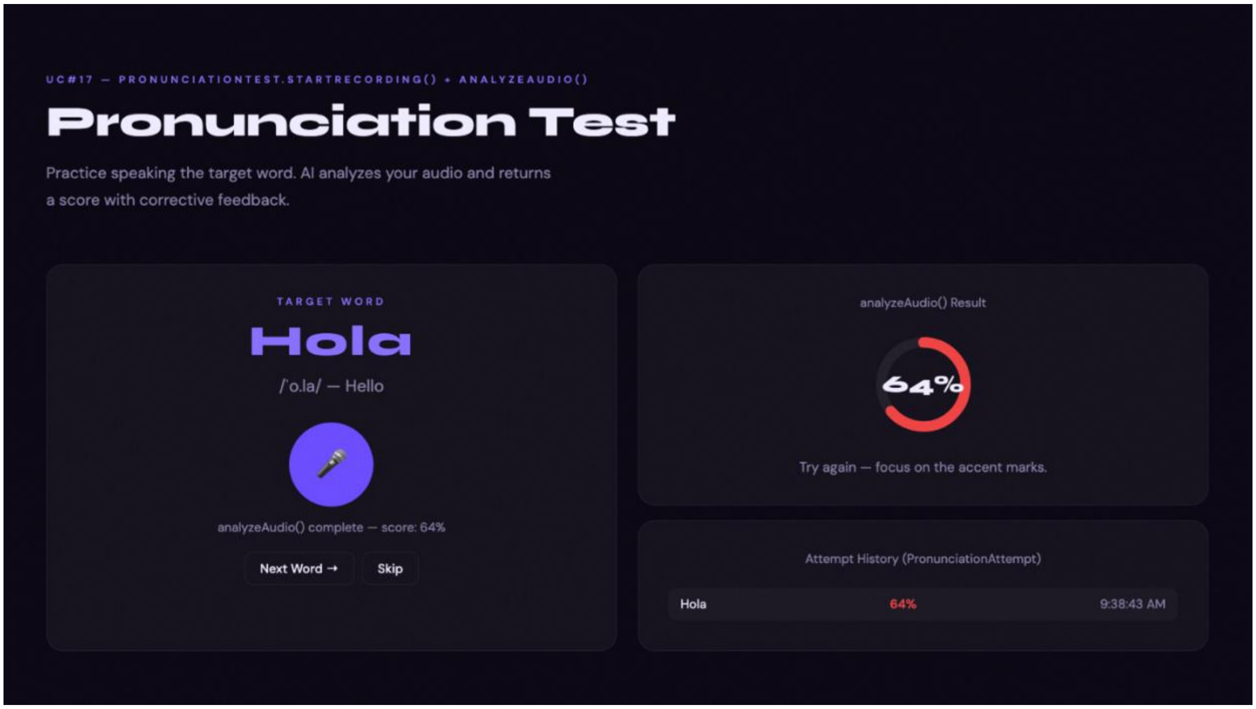

UC#020 — LEARNINGPATH.ASSIGNLEVEL()
Select Your Target Language
Choose a language to initialize your LearningPath. After confirming, you'll be directed to the Pre-Test.
Spanish
Español
French
Français
Japanese
日本語
Mandarin
中文
German
Deutsch
Portuguese
Português
Korean
한국어
Italian
Italiano
Confirm Selection →

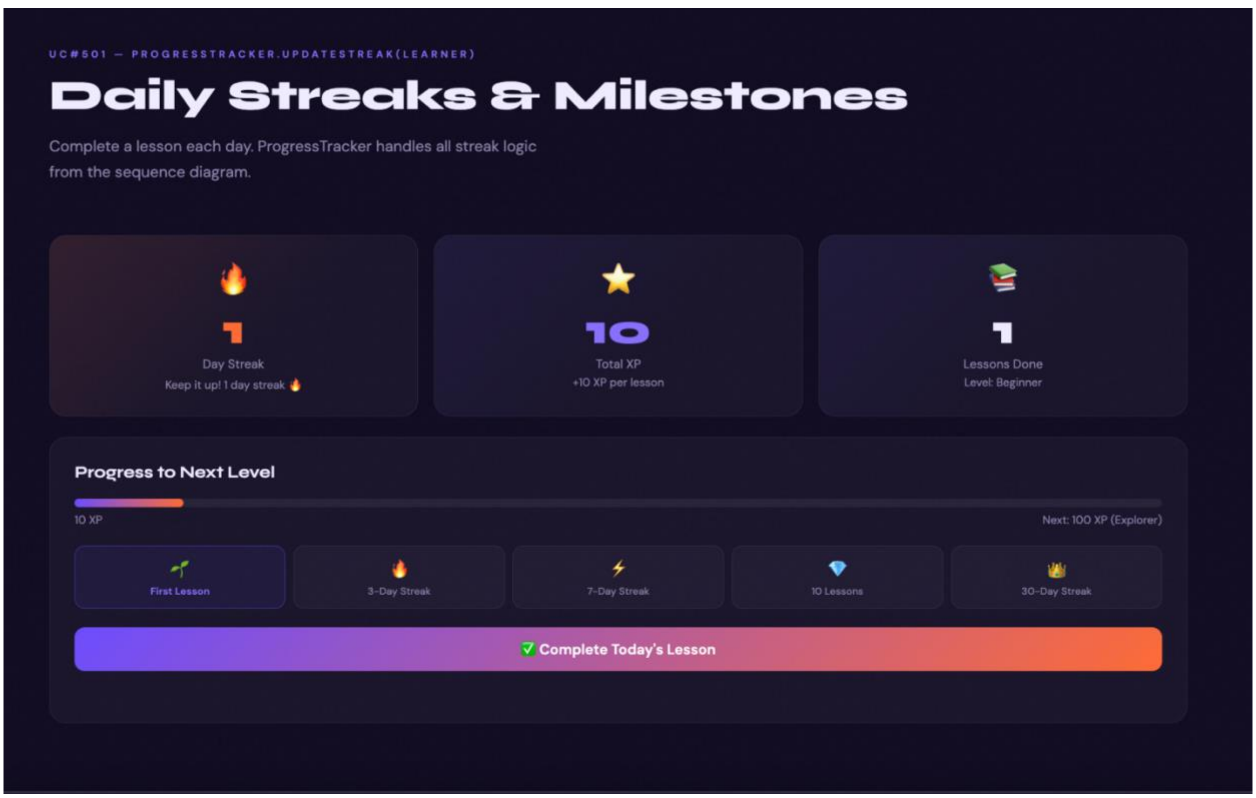

UC#562 — QUIZ.EVALUATEANSWERS() → AI PLACEMENT
Skill Assessment Pre-Test
Answer 5 questions. AI scores your results and assigns your starting level. You can skip to default Beginner.
QUESTION 1 OF 5 · SPANISH
What does "Hola" mean?
Goodbye
Hello
Thank you
Please
Skip Pre-Test → Default Beginner

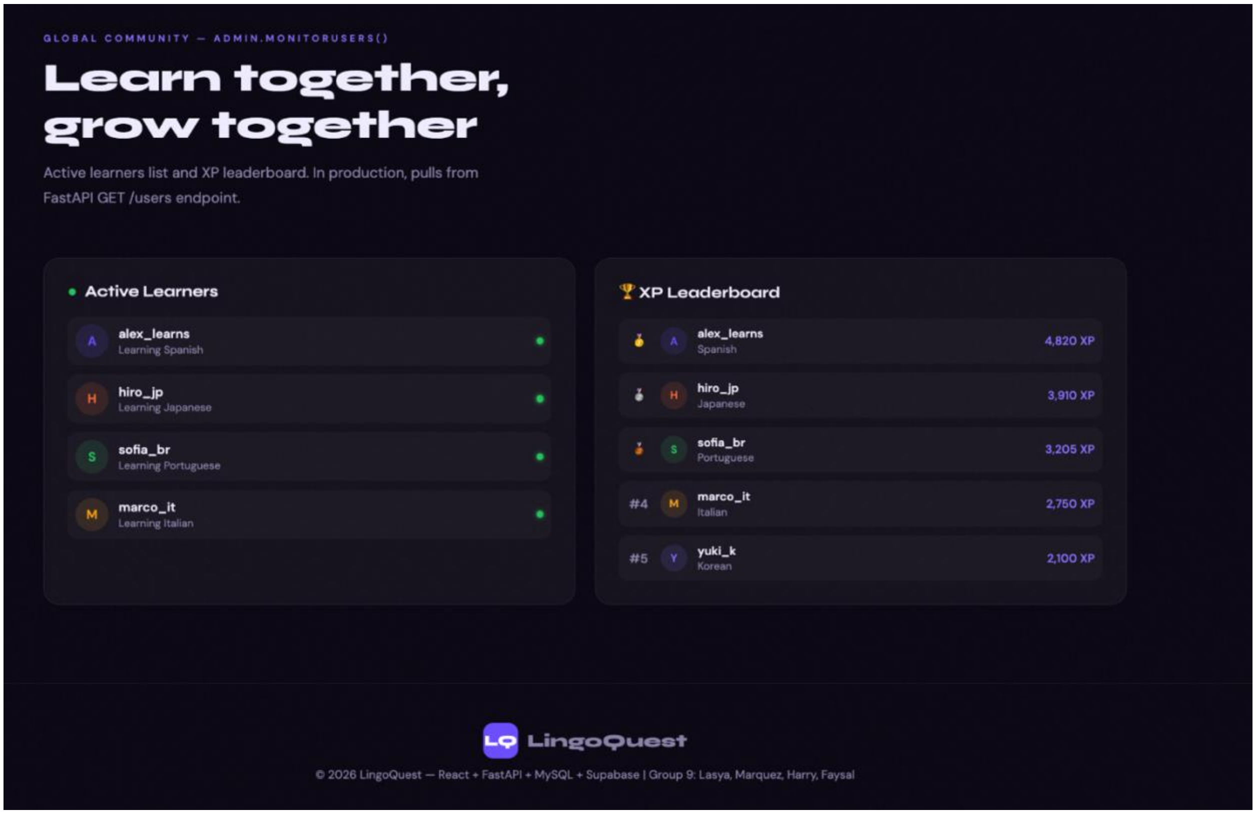

UC#24 — LESSON.LOADLESSON() + MARKCOMPLETE(LEARNERID)
Level-Based Lessons
Select a lesson from your assigned level. Complete it to update your progress and unlock the quiz.
Greetings & Introductions
Beginner
Numbers 1–10
Beginner
Colors & Adjectives
Intermediate
Food & Dining
Intermediate
Greetings & Introductions
Beginner
In this lesson you'll learn basic Spanish greetings. These are the most common words you'll use every day.
VOCABULARY — LANGUAGE.LOADLESSONS()
Hola
Hello
Adiós
Goodbye
Gracias
Thank you
Por favor
Please
markComplete(learnerId)


UC#645 — QUIZ.ADJUSTDIFFICULTY(SCORE): VOID
Adaptive Quiz
Questions get harder or easier based on your accuracy. Score above threshold to unlock the next level.
Easy
QUESTION 1/3 · 0 CORRECT
What does "Hola" mean?
Hello
Goodbye
Thanks
Sorry
Session Accuracy
0%


UC#966 — SKETCHPADNOTE.SAVENOTE(DATA: BYTE[])
Visual Sketchpad
Draw notes linked to your current lesson. Saved as SketchpadNote with drawingData and savedAt timestamp.
Color:
Size:
Erase
SWE 26
saveNote(data)
clearNote()
Export
clearNote() called — canvas reset


UC#17 — PRONUNCIATIONTEST.STARTRECORDING() + ANALYZEAUDIO()
Pronunciation Test
Practice speaking the target word. AI analyzes your audio and returns a score with corrective feedback.
TARGET WORD
Hola
/ˈo.la/ — Hello
analyzeAudio() complete — score: 64%
Next Word →
Skip
analyzeAudio() Result
64%
Try again — focus on the accent marks.
Attempt History (PronunciationAttempt)
Hola
64%
9:38:43 AM


UC#S01 — PROGRESSTRACKER.UPDATESTREAK(LEARNER)
Daily Streaks & Milestones
Complete a lesson each day. ProgressTracker handles all streak logic from the sequence diagram.
1
Day Streak
Keep it up! 1 day streak
10
Total XP
+10 XP per lesson
1
Lessons Done
Level: Beginner
Progress to Next Level
10 XP
Next: 100 XP (Explorer)
First Lesson
3-Day Streak
7-Day Streak
10 Lessons
30-Day Streak
Complete Today's Lesson



4.2 Test Cases

Test Case ID

TC-01

Test Case Name- User Registration with Valid Data

Precondition: User is on the registration page

Test Steps:

-Enter Valid First Name, Last Name, Email, and Password
- Click Sign Up

Expected Result:

The account is created successfully, the user profile is stored in the database, and the user is redirected to the dashboard.

TC-02

Test Case Name- User Login with Invalid Credentials

Precondition: User account exists/does not exist

Test Steps:
-Open Login Page
- Enter Incorrect Email/Password
-Click Log in

Expected Result:

The system denies access, shows an error message for invalid credentials, and allows retry.

TC-03

Test Case Name- Select Target Language

Precondition: User is logged in

Test Steps:
-Open Language selection page
- Choose a language
-Confirm Selection

Expected Result:

The system saves the selected language, then initializes the learning path, and redirects the user to the pre-test or first lesson.

TC-04

Test Case Name- Skip Skill Pre-Test

Precondition: User is logged in with selected language

Test Steps:
-Open Pre Test Page
- Click Skip Pre Test

Expected Result:

The system assigns the user to the Beginner level by default and redirects to the starting lesson.

TC-05

Test Case Name- Complete Lesson Successfully

Precondition: User has an assigned level and lesson content is available

Test Steps:
-Select a lesson
- Review Lesson Pages and examples
-Complete Lesson

Expected Result:

Lessons are marked complete, progress is updated, and completion status is saved.

TC-06

Test Case Name- Adaptive Quiz with Correct Answers

Precondition: User is in an active learning path and a quiz is available

Test Steps:
-Start Quiz
- Submit Correct Answers
-Finish quiz

Expected Result:

The system calculates scores, stores quiz results, updates progress and unlocks the next level.

TC-07

Test Case Name- Save Sketchpad Notes

Precondition: User is logged in and viewing a lesson.

Test Steps:
-Open Sketchpad
-Draw or write notes
-Save
-Reopen same pad

Expected Result:

The notes are stored and tied to that lesson and user account, saved notes appear again when reopened.

TC-08

Test Case Name- Microphone Pronunciation Test

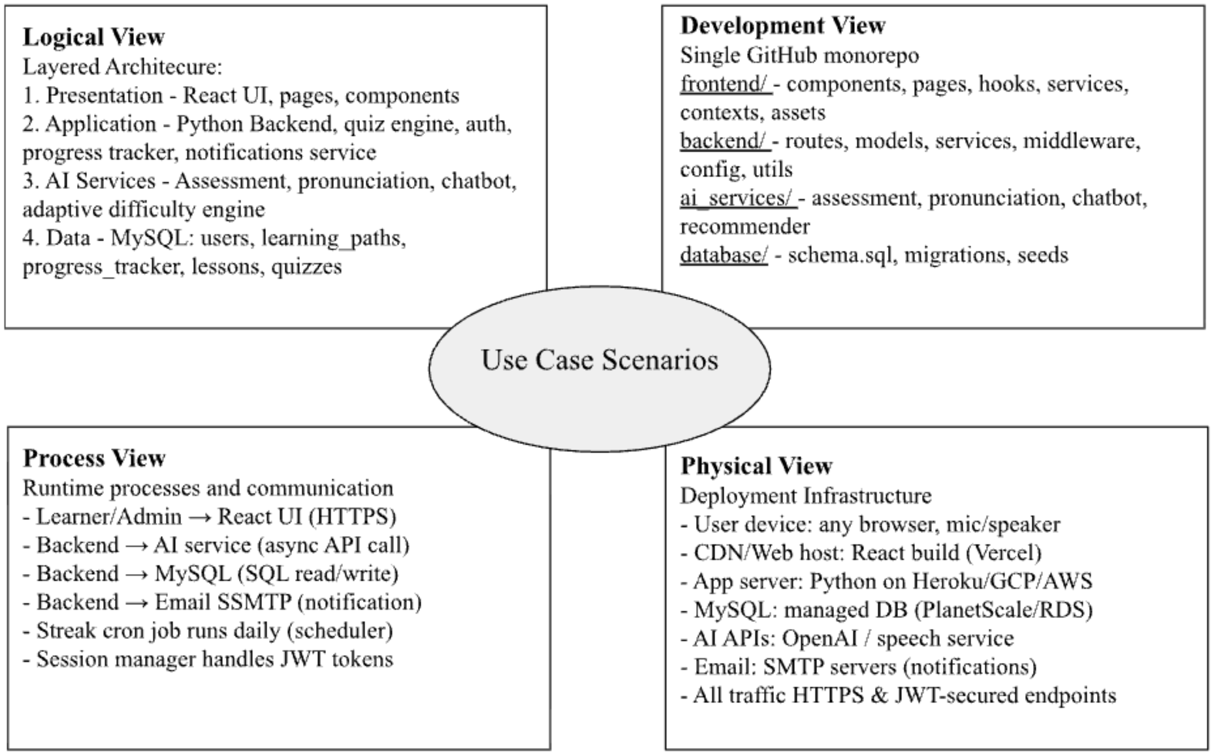
Precondition: User is logged in and has microphone access

Test Steps:
-Open verbal exercise
- Record and Submit Audio

Expected Result:

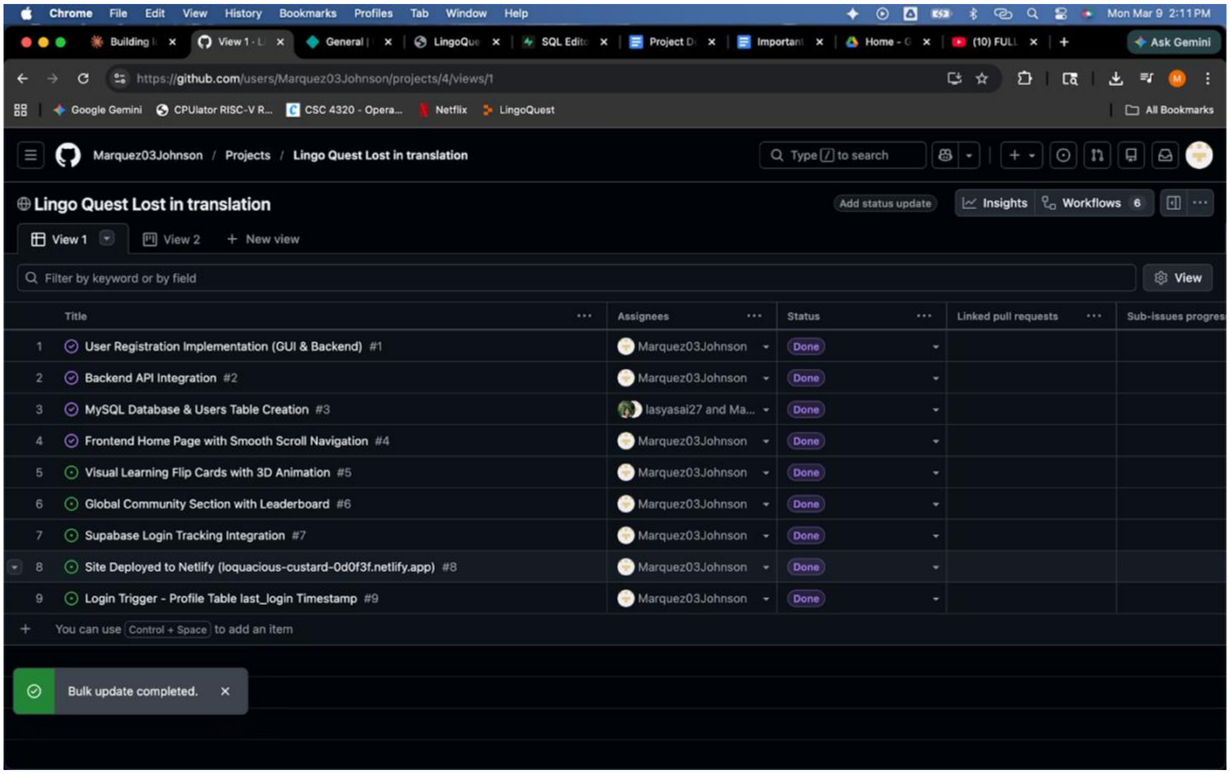
The system sends the audio to verify then displays pronunciation score and stores attempt history.

4.3 Architectural Modeling



4.4 Git Hub Links & Screenshots

Link:


Marquez03Johnson / Projects / Lingo Quest Lost in translation
Lingo Quest Lost in translation
Add status update
Insights
Workflows 6
View 1
View 2
New view
Filter by keyword or by field
View
Title
Assignees
Status
Linked pull requests
Sub-issues progress
User Registration Implementation (GUI & Backend) #1
Backend API Integration #2
MySQL Database & Users Table Creation #3
Frontend Home Page with Smooth Scroll Navigation #4
Visual Learning Flip Cards with 3D Animation #5
Global Community Section with Leaderboard #6
Supabase Login Tracking Integration #7
Site Deployed to Netlify (loquacious-custard-0d0f3f.netlify.app) #8
Login Trigger - Profile Table last_login Timestamp #9
Marquez03Johnson
lasyasai27 and Ma...
Done
You can use Control + Space to add an item
Bulk update completed.